NORTHROP

Northrop Hardhat

User Guide

For engineers: how to run an inspection, from loading drawings to sending the memo.

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Built by David Samson

Internal document: not for clients.

Getting started

Install it on the iPad

1. Open the app's link in **Safari** on the iPad.
2. Tap the **Share** button, then **Add to Home Screen**.
3. Open it from the home screen from now on. It runs full screen and works with no internet.

> When an update is ready, the app shows **Update now**. Tap it, or close and reopen the app.

Set up your details

1. Open **Settings** from the bar on the left.
2. Under **My details**, enter your name and title. They go on new inspections and memos.
3. Draw or upload your **signature**. It's copied into each new memo.

Keep your work safe

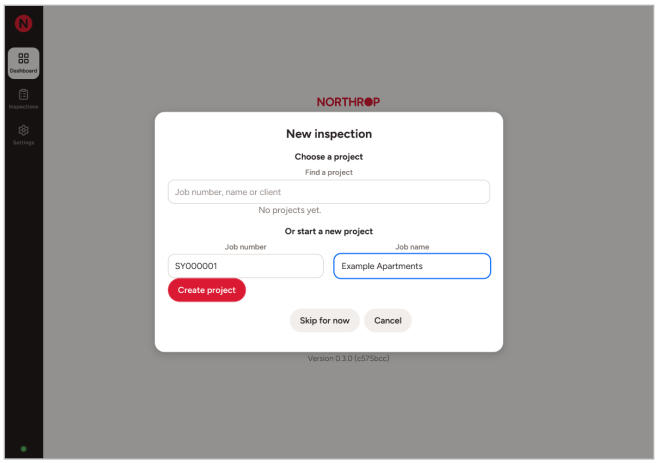
Your work is stored only on this iPad, and iOS can clear a web app's storage. **Back up each inspection** when you finish it (see Backing up), and use **Back up all** in Settings now and then.

> **Never remove Hardhat from the home screen.** Removing it deletes every inspection, drawing and photo in it. If you ever have to, use **Back up all** first and keep the files somewhere safe.

Before you go to site

Start an inspection

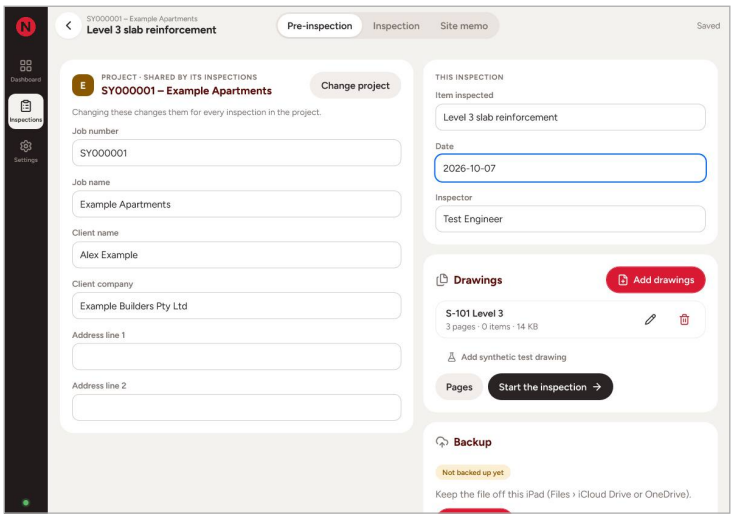
1. On the **Dashboard**, tap **New inspection**.
2. Choose the project, or start a new one with its job number and job name. (You can **Skip for now** and assign a project later.)
3. The inspection opens on **Pre-inspection**.



New inspection: choose a project or start a new one

Fill in the details and load drawings

1. Check the project details (client, address) and fill in **Item inspected**, for example "Level 3 slab reinforcement". Everything saves as you type.
2. In **Drawings**, tap **Add drawings** and pick the PDFs from Files. Drag the handle to put them in order.
3. Tap **Start the inspection** when you're ready.



Pre-inspection: the project, the inspection details and the drawings

On site

Getting around the drawings

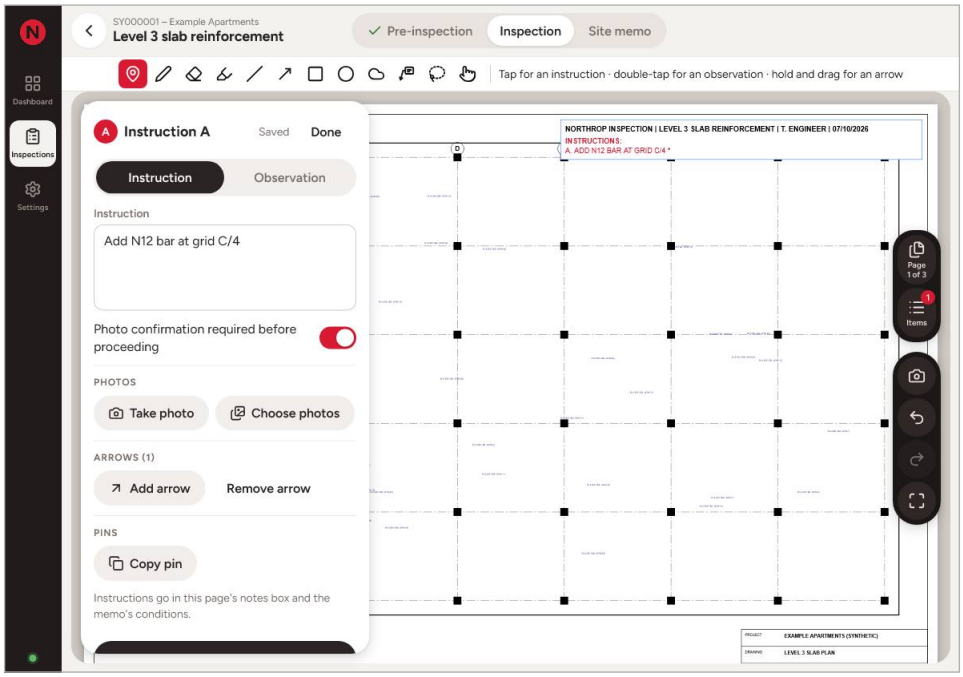
- All the drawings are one long document. **Scroll with one finger**, **pinch with two** to zoom.
- The **Page** button on the right opens the Pages view: tap a page to go to it. From each page's menu you can duplicate, hide or move it. Hold a page, then drag it to move it anywhere, even into another drawing. **Include in export** puts a page without pins in the PDF pack, and **Add sketch page after…** inserts a Northrop calc pad sheet (A4–A1, portrait or landscape, with a scale) with a light grid to sketch on; it's filled in from the job and exported with the rest.
- **Fit page** (bottom right) fits the current page to the screen.
- The Apple Pencil never scrolls: it's for pins and drawing.

Pins: instructions and observations

Turn on the **Pin** tool (first in the toolbar), then:

- **Tap** for an **instruction** (red pin).
- **Double-tap** for an **observation** (blue pin), or to switch an existing pin's kind.
- **Tap and hold on the spot you mean, then drag** for a pin with an **arrow**: the arrowhead stays on the spot and the pin follows your finger to a clear place.

The item's editor opens: type the instruction or observation. For an instruction the builder must prove, turn on **Photo confirmation required**. Tap **Done**, or tap the drawing to close it.



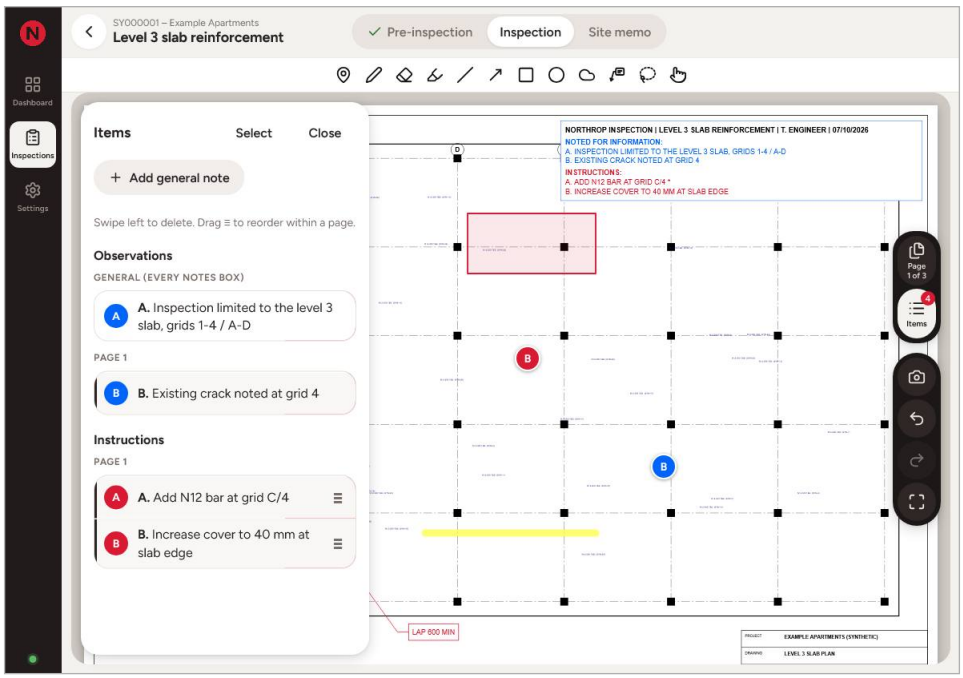
An instruction's editor

- Tap a pin (with any tool) to open it again; drag it to move it.
- **Add arrow** points the pin at more spots; **Copy pin** puts the same item at another spot.

- Letters are automatic: instructions A, B, C… and observations A, B, C… in order through the drawings.
- The **notes box** on each page lists its items. Drag it somewhere clear: it's printed on the drawing. Tap it to change its width (drag a side handle) or its text size (**S**, **M**, **L**); Undo reverses moves and size changes.

The Items panel

Tap **Items** on the right to list every item by page. Tap one to jump to it, swipe left to delete, drag the handle to reorder items on a page, or tap **Select** to change several at once.



The Items panel

General notes

For something about the whole inspection rather than one spot, such as "Inspection limited to the roof framing, grids 1–4 / A–C":

1. Open **Items** and tap **Add general note**.
2. Type the note and tap **Done**.

General notes are observations with no pin. They're lettered first (A, B…), before the pinned observations, and every page's notes box lists them. Edit or delete them from the Items panel like any item.

On Pre-inspection, the **Inspection type** buttons (General concrete, piling, steel…) add each type's standard general notes in one go; tapping a type again takes them away (it asks first if you edited one). The types and their notes are in Settings, Prefilled messages, **Inspection types**.

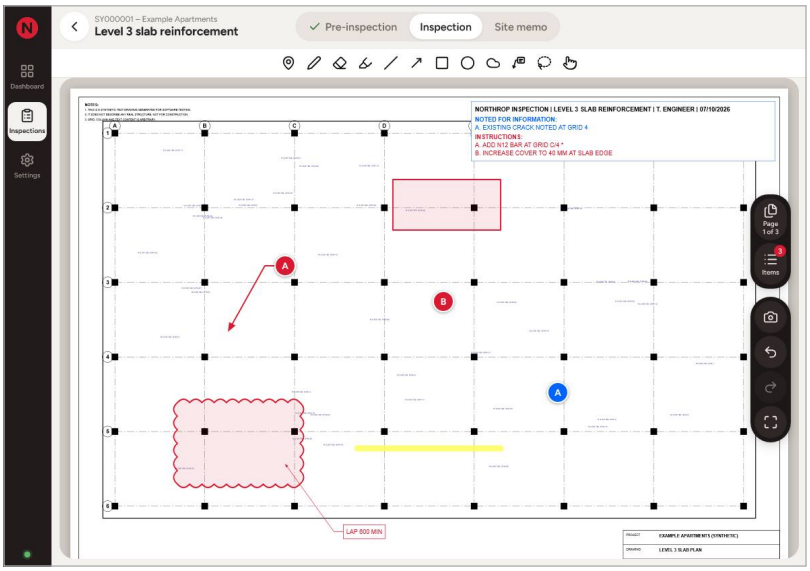
Photos

- In an item's editor: **Take photo** (camera) or **Choose photos** (library).
- The **camera** button on the right takes a general photo, not tied to a pin.
- Tap a photo to view it full screen and add a caption.
- **Save to iPad** shares the photos to Photos or Files for filing.

Marking up with the Pencil

Each tool is a toggle: tap it to turn it on, tap again to turn it off. With a drawing tool on, the **Pencil draws** and a **finger scrolls**. (No Pencil? Turn on **Draw with finger**: one finger draws, two scroll.)

- **Pen** and **Highlighter:** pick a weight and a colour slot. Tap the colour in use again for more colours.
- **Shapes:** line, arrow, rectangle, ellipse, revision cloud. Drag from corner to corner.
- **Draw and hold:** draw a shape freehand and hold the Pencil still at the end: it snaps to a clean line, circle, rectangle, triangle or polygon. Keep pressing and drag to make it bigger or smaller.
- **Eraser:** rub over a mark to remove it. Tap inside a filled shape to take its fill off.
- **Text:** tap for a text box, or tap and hold on the spot you mean, then drag the box out to a clear place: its arrow points back to the spot. Type in capitals as you like; tap elsewhere to finish.



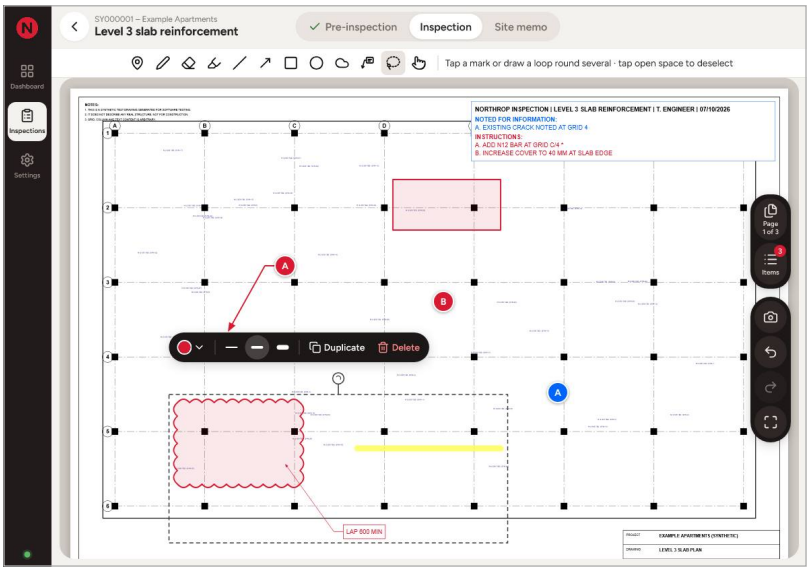
Markup on the drawing

Select

Turn on **Select** (the dotted lasso), then tap a mark or draw a loop round several. Quicker still: with no tool on, just **tap any mark you've drawn**. Select comes on with it picked, and tapping open space lets go. (With the Pen or another drawing tool on, a tap always draws, so writing never picks up a nearby stroke.)

- **Drag** inside the dashed box to move them.
- The bar above changes the **colour** or **weight**, **duplicates** or **deletes** them.
- One shape shows **handles** to resize it. Tap a selected callout again to edit its text.

- Drag the **round handle** above the selection to **rotate** it (one shape, or a group together). It snaps to level and 45°.
- Tap open space to let go.



Two marks selected, with the bar

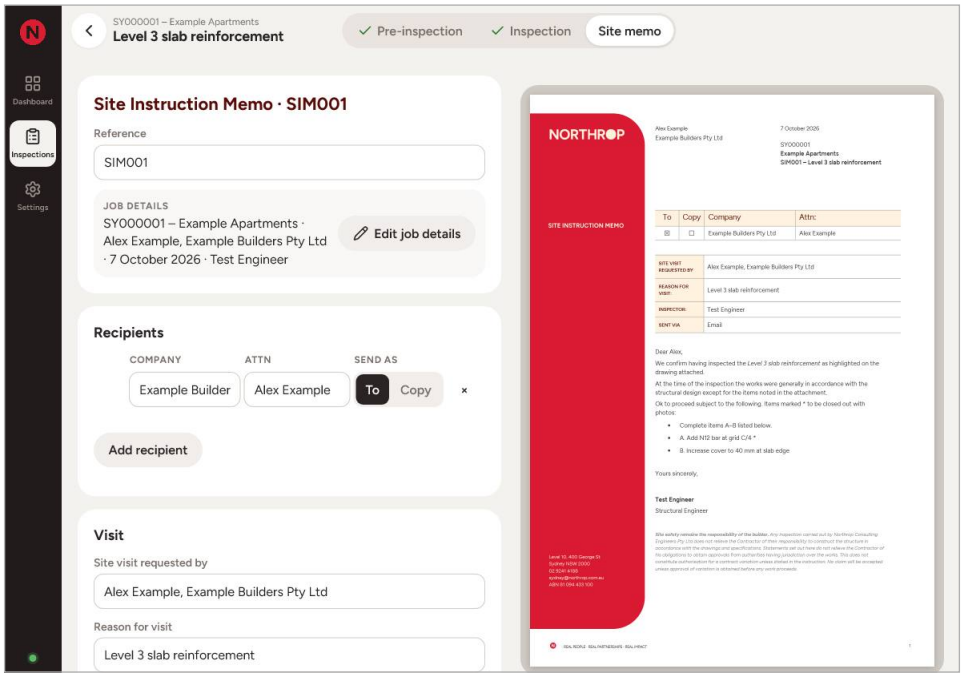
Undo

- The **Undo** and **Redo** buttons on the right undo pins, deletes, markup and more.
- Or **tap twice with two fingers** anywhere on the drawing to undo.

After the inspection

The Site Instruction Memo

1. Open **Site memo** and tap **Create memo**. It gets the job's next reference, such as SIM001 (a project can use other letters: set **Memo reference letters** on the project's page).
2. Check the **recipients**: the client starts as "To". Add others as To or Copy, and drag ⠿ to change their order.
3. Pick the **message** that fits, and tick the **standard conditions** that apply. Your instructions are listed as conditions; each can be reworded for the memo. Instructions needing photos are marked *, and the line above the list explains it. With lots of pins, turn on **Refer to drawings**: one line points the builder to the attached drawings instead of listing every instruction.
4. Check the **signature** and sign-off.
5. Read the **preview** beside the editor: it's exactly what will be exported.



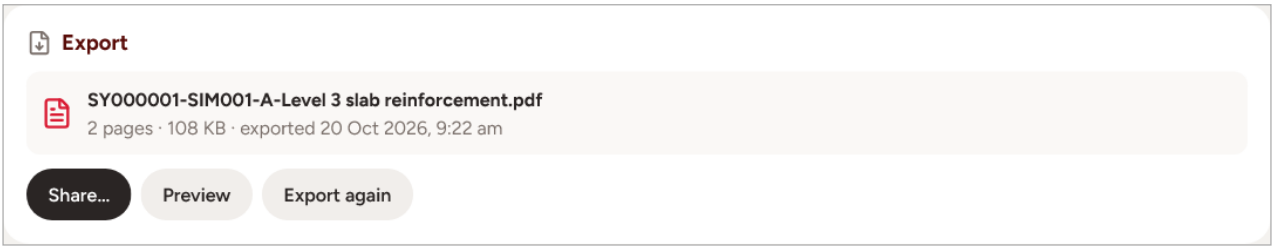
The memo editor and its preview

Export the PDF pack

1. At the end of Site memo, tap **Export PDF**.
2. When it's ready, tap **Share…** to send it by Mail, save it to Files or upload it to Aconex.
3. The pack holds the memo, the marked-up drawing pages and the photo appendix.

Under it, **Export Word** makes the memo and photo appendix as an editable Word document (with the Northrop sidebar and fonts), and **Export defect register** makes an Excel file with one row per item (references, descriptions, sheets, photo references) and empty Status, Date closed and Comments columns for the builder; **Share register…** sends it.

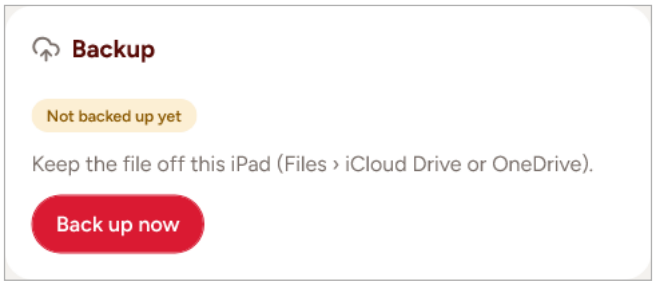
> If you change pins after exporting, the letters may change. The app warns you and lists what changed: export again before sending.



The Export card, ready to share

Backing up

1. Open **Pre-inspection** and find the **Backup** card.
2. Tap **Back up now**, then **Share…** and choose **Save to Files** (or AirDrop it to your computer).
3. To restore or move it to another device, use **Import inspection…** on the Inspections page.



The Backup card

Gesture cheat sheet

Fingers

- **One finger:** scroll. **Two fingers:** pinch to zoom.
- **Two-finger double-tap:** undo.
- **Tap a pin or callout:** open it. **Drag it:** move it.

Pin tool

- **Tap:** instruction pin.
- **Double-tap:** observation pin (or switch a pin's kind).
- **Tap, hold, drag:** pin with an arrow (hold on the spot, drag the pin out).
- **Double-tap, hold, drag:** observation with an arrow (the same way).

Pencil

- **Draw and hold:** straight line, or a clean shape from a closed stroke; keep dragging to resize it.
- **Text tool:** tap for a box; tap, hold on the spot and drag the box out for one with an arrow.
- **Select tool:** tap a mark, or loop round several; drag inside the box to move; drag the round handle to rotate.
- **Eraser:** rub to remove; tap inside a shape to take its fill off.